

CN NS2 WIRED SIMLULATION

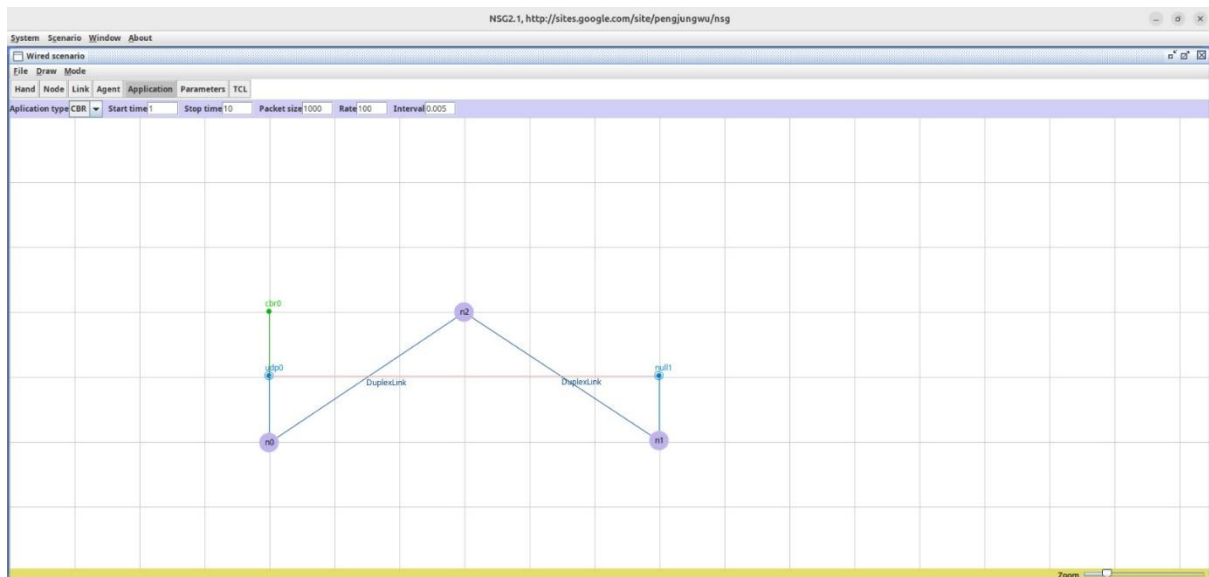
NAME: Rithvik Rajesh Matta

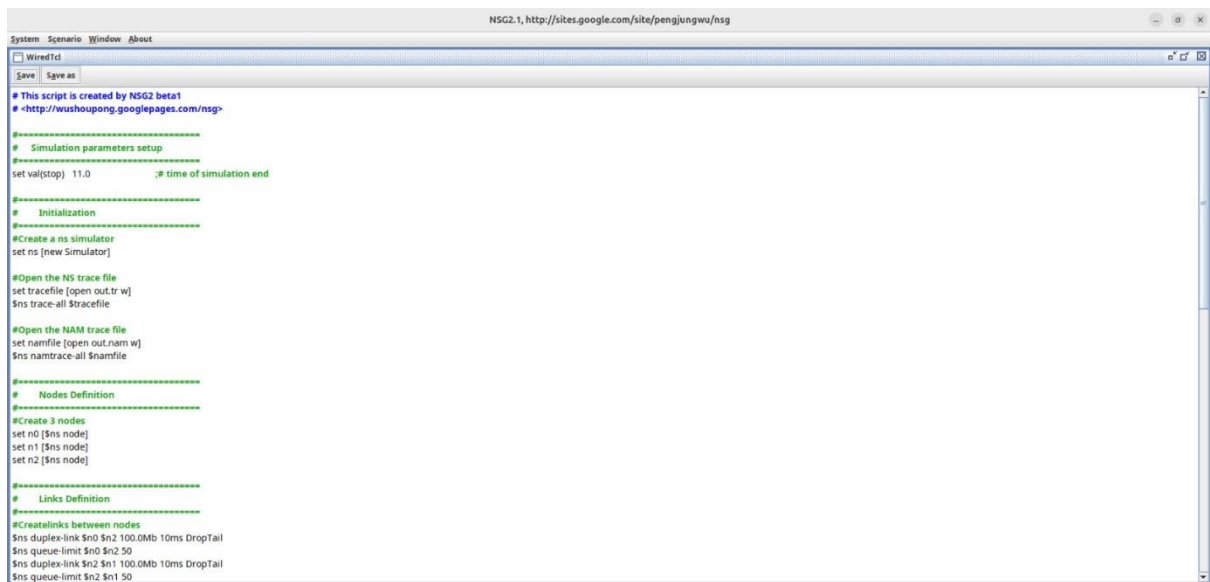
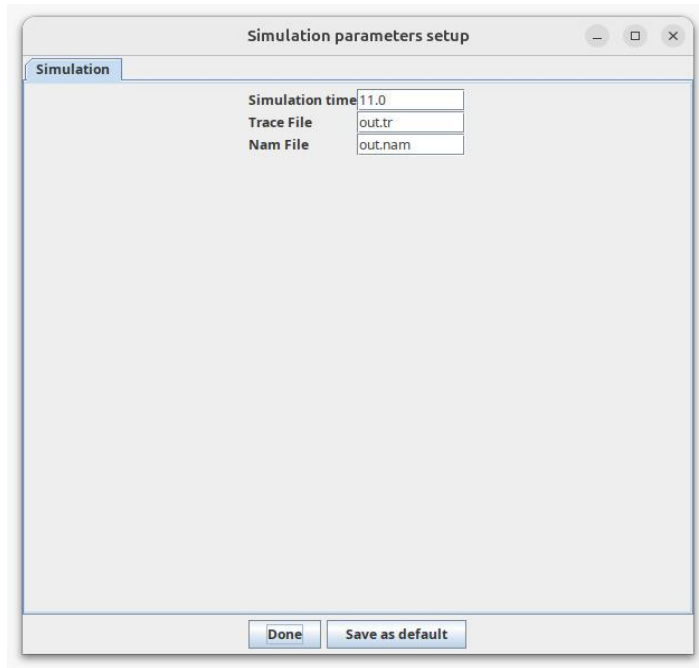
SRN: PES2UG23CS485

CLASS: 4H-CSE

DATE: 25/03/2025

NSG2.1





```

System Scenario Window About
NSG2.1, http://sites.google.com/site/pengjungwu/nsg
WiredTcl
Save Save as

#-----
# Links Definition
#-----
# Create links between nodes
$ns duplex-link $n0 $n2 100.0Mb 10ms DropTail
$ns queue-limit $n0 $n2 50
$ns duplex-link $n2 $n1 100.0Mb 10ms DropTail
$ns queue-limit $n2 $n1 50

# Give node position (for NAM)
$ns duplex-link-op $n0 $n2 orient right-up
$ns duplex-link-op $n2 $n1 orient right-down

#-----
# Agents Definition
#-----
# Setup a UDP connection
set udp0 [new Agent/UDP]
$ns attach-agent $n0 $udp0
set null1 [new Agent/Null]
$ns attach-agent $n1 $null1
$ns connect $udp0 $null1
$udp0 set packetSize_1500

#-----
# Applications Definition
#-----
# Setup a CBR Application over UDP connection
set cbr0 [new Application/Traffic/CBR]
$cbr0 attach-agent $udp0
$cbr0 set packetSize_1000
$cbr0 set rate_100.0Mb
$cbr0 set random_null
$ns at 1.0 "$cbr0 start"
$ns at 10.0 "$cbr0 stop"

```

```

System Scenario Window About
NSG2.1, http://sites.google.com/site/pengjungwu/nsg
WiredTcl
Save Save as

set udp0 [new Agent/UDP]
$ns attach-agent $n0 $udp0
set null1 [new Agent/Null]
$ns attach-agent $n1 $null1
$ns connect $udp0 $null1
$udp0 set packetSize_1500

#-----
# Applications Definition
#-----
# Setup a CBR Application over UDP connection
set cbr0 [new Application/Traffic/CBR]
$cbr0 attach-agent $udp0
$cbr0 set packetSize_1000
$cbr0 set rate_100.0Mb
$cbr0 set random_null
$ns at 1.0 "$cbr0 start"
$ns at 10.0 "$cbr0 stop"

#-----
# Termination
#-----
# Define a "finish" procedure
proc finish {} {
    global ns tracefile namfile
    $ns flush-trace
    close $tracefile
    close $namfile
    exec nam out.nam &
    exit 0
}
$ns at $val(stop) "$ns nam-end-wireless $val(stop)"
$ns at $val(stop) "finish"
$ns at $val(stop) "puts 'done!'; $ns halt"
$ns run

```

nano prog.tcl

```
GNU nano 7.2                                prog.tcl
# This script is created by NSG2 beta1
# <http://wushoupong.googlepages.com/nsg2>

# Simulation parameters setup
#####
set val(stop) 11.0                          ;# time of simulation end
#####

# Initialization
#####
#Create a ns simulator
set ns [new Simulator]

#Open the NS trace file
set tracefile [open out.tr w]
$ns trace-all $tracefile

#Open the NAM trace file
set namfile [open out.nam w]
$ns namtrace-all $namfile
#####

# Nodes Definition
#####
#Create 3 nodes
set n0 [$ns node]
set n1 [$ns node]
set n2 [$ns node]

# Links Definition
#####
#Createlinks between nodes
#####
```

```
#####
# Links Definition
#####
#Createlinks between nodes
$ns duplex-link $n0 $n2 100.0Mb 10ms DropTail
$ns queue-limit $n0 $n2 50
$ns duplex-link $n2 $n1 99.9Mb 10ms DropTail
$ns queue-limit $n2 $n1 50

#Give node position (for NAM)
$ns duplex-link-op $n0 $n2 orient right-up
$ns duplex-link-op $n2 $n1 orient right-down
#####

# Agents Definition
#####
#Setup a UDP connection
set udp0 [new Agent/UDP]
$ns attach-agent $n0 $udp0
set null1 [new Agent/Null]
$ns attach-agent $n1 $null1
$ns connect $udp0 $null1
$udp0 set packetSize_ 1500

# Applications Definition
#####
#Setup a CBR Application over UDP connection
set cbr0 [new Application/Traffic/CBR]
$cbr0 attach-agent $udp0
$cbr0 set packetSize_ 1000
$cbr0 set rate_ 100.0Mb
$cbr0 set random_ null
```

```
#####
$ns connect $udp0 $null1
$udp0 set packetSize_ 1500

# Applications Definition
#####
#Setup a CBR Application over UDP connection
set cbr0 [new Application/Traffic/CBR]
$cbr0 attach-agent $udp0
$cbr0 set packetSize_ 1000
$cbr0 set rate_ 100.0Mb
$cbr0 set random_ null
$ns at 1.0 "$cbr0 start"
$ns at 10.0 "$cbr0 stop"

# Termination
#####
#Define a 'finish' procedure
proc finish {} {
    global ns tracefile namfile
    $ns flush-trace
    close $tracefile
    close $namfile
    exec nam out.nam &
    exit 0
}
$ns at $val(stop) "$ns nam-end-wireless $val(stop)"
$ns at $val(stop) "finish"
$ns at $val(stop) "puts \"done\" ; $ns halt"
$ns run
```

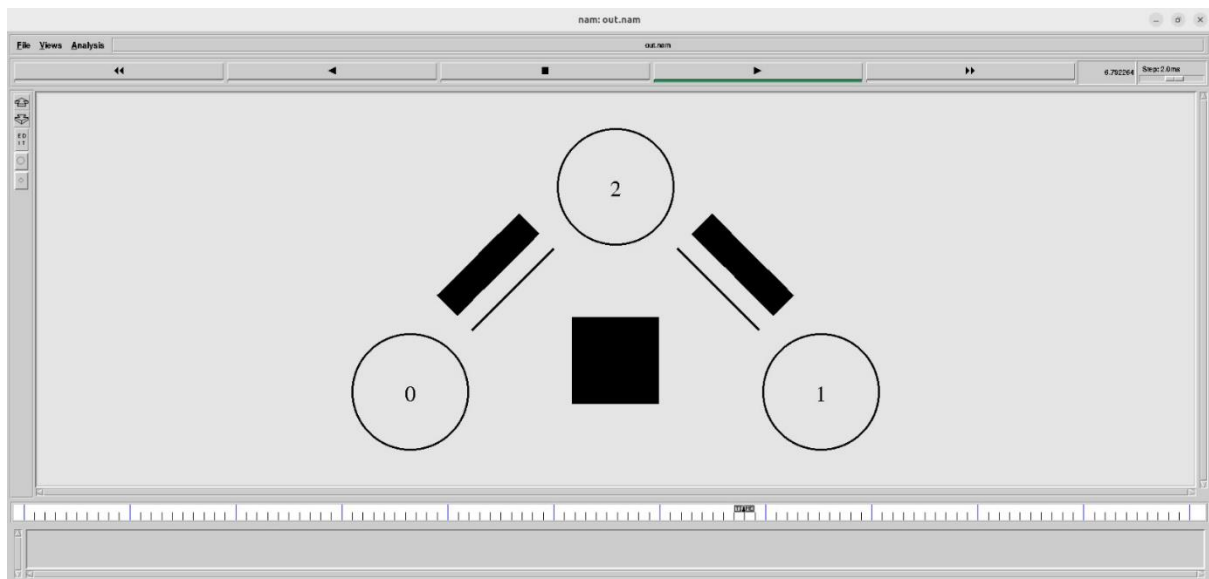
nano out.tr

```
GNU nano 7.2 out.tr
1 0 2 cbr 1000 ----- 0 0.0 1.0 0 0
1 0 2 cbr 1000 ----- 0 0.0 1.0 0 0
1.00088 0 2 cbr 1000 ----- 0 0.0 1.0 1 1
1.00088 0 2 cbr 1000 ----- 0 0.0 1.0 1 1
1.00016 0 2 cbr 1000 ----- 0 0.0 1.0 2 2
1.00016 0 2 cbr 1000 ----- 0 0.0 1.0 2 2
1.00024 0 2 cbr 1000 ----- 0 0.0 1.0 3 3
1.00024 0 2 cbr 1000 ----- 0 0.0 1.0 3 3
1.00032 0 2 cbr 1000 ----- 0 0.0 1.0 4 4
1.00032 0 2 cbr 1000 ----- 0 0.0 1.0 4 4
1.0004 0 2 cbr 1000 ----- 0 0.0 1.0 5 5
1.0004 0 2 cbr 1000 ----- 0 0.0 1.0 5 5
1.00048 0 2 cbr 1000 ----- 0 0.0 1.0 6 6
1.00048 0 2 cbr 1000 ----- 0 0.0 1.0 6 6
1.00056 0 2 cbr 1000 ----- 0 0.0 1.0 7 7
1.00056 0 2 cbr 1000 ----- 0 0.0 1.0 7 7
1.00064 0 2 cbr 1000 ----- 0 0.0 1.0 8 8
1.00064 0 2 cbr 1000 ----- 0 0.0 1.0 8 8
1.00072 0 2 cbr 1000 ----- 0 0.0 1.0 9 9
1.00072 0 2 cbr 1000 ----- 0 0.0 1.0 9 9
1.0008 0 2 cbr 1000 ----- 0 0.0 1.0 10 10
1.0008 0 2 cbr 1000 ----- 0 0.0 1.0 10 10
1.00088 0 2 cbr 1000 ----- 0 0.0 1.0 11 11
1.00088 0 2 cbr 1000 ----- 0 0.0 1.0 11 11
1.00096 0 2 cbr 1000 ----- 0 0.0 1.0 12 12
1.00096 0 2 cbr 1000 ----- 0 0.0 1.0 12 12
1.00104 0 2 cbr 1000 ----- 0 0.0 1.0 13 13
1.00104 0 2 cbr 1000 ----- 0 0.0 1.0 13 13
1.00112 0 2 cbr 1000 ----- 0 0.0 1.0 14 14
1.00112 0 2 cbr 1000 ----- 0 0.0 1.0 14 14
1.0012 0 2 cbr 1000 ----- 0 0.0 1.0 15 15
1.0012 0 2 cbr 1000 ----- 0 0.0 1.0 15 15
1.00128 0 2 cbr 1000 ----- 0 0.0 1.0 16 16
1.00128 0 2 cbr 1000 ----- 0 0.0 1.0 16 16
[ Read 674942 lines ]
^C Help      ^O Write Out  ^N Where Is   ^K Cut        ^T Execute    ^C Location   ^U Undo       ^M-A Set Mark  ^I To Bracket ^M-C Previous ^B Back
^X Exit      ^R Read File  ^X Replace    ^J Paste      ^_ Justify    ^G Go To Line ^E Redo       ^M-d Copy      ^G Where Mas  ^M-N Next     ^F Forward
```

nano out.nam

```
GNU nano 7.2 out.nam
^t * -v 1.0a5 -a 0
A ^t * -n 1 -p 0 -o 0x7fffffff -c 30 -a 1
A ^t * -h 1 -n 1073741023 -s 0
n ^t * -a 0 -s 0 -5 UP -v circle -c black -t black
n ^t * -a 1 -s 1 -5 UP -v circle -c black -t black
n ^t * -a 2 -s 2 -5 UP -v circle -c black -t black
l ^t * -s 0 -d 2 -5 UP -r 100000000 -D 0.01 -c black -o right-up
l ^t * -s 2 -d 1 -5 UP -r 999000000 -D 0.01 -c black -o right-down
^t 1 -s 0 -d 2 -p cbr -e 1000 -c 0 -i 0 -a 0 -x {0.0 1.0 0 ----- null}
^t 1 -s 0 -d 2 -p cbr -e 1000 -c 0 -i 0 -a 0 -x {0.0 1.0 0 ----- null}
h ^t 1 -s 0 -d 2 -p cbr -e 1000 -c 0 -i 0 -a 0 -x {0.0 1.0 1 ----- null}
^t 1.00008 -s 0 -d 2 -p cbr -e 1000 -c 0 -i 1 -a 0 -x {0.0 1.0 1 ----- null}
h ^t 1.00008 -s 0 -d 2 -p cbr -e 1000 -c 0 -i 1 -a 0 -x {0.0 1.0 1 ----- null}
^t 1.00016 -s 0 -d 2 -p cbr -e 1000 -c 0 -i 2 -a 0 -x {0.0 1.0 2 ----- null}
h ^t 1.00016 -s 0 -d 2 -p cbr -e 1000 -c 0 -i 2 -a 0 -x {0.0 1.0 2 ----- null}
^t 1.00016 -s 0 -d 2 -p cbr -e 1000 -c 0 -i 2 -a 0 -x {0.0 1.0 2 ----- null}
h ^t 1.00016 -s 0 -d 2 -p cbr -e 1000 -c 0 -i 2 -a 0 -x {0.0 1.0 2 ----- null}
^t 1.00024 -s 0 -d 2 -p cbr -e 1000 -c 0 -i 3 -a 0 -x {0.0 1.0 3 ----- null}
h ^t 1.00024 -s 0 -d 2 -p cbr -e 1000 -c 0 -i 3 -a 0 -x {0.0 1.0 3 ----- null}
^t 1.00024 -s 0 -d 2 -p cbr -e 1000 -c 0 -i 3 -a 0 -x {0.0 1.0 3 ----- null}
h ^t 1.00024 -s 0 -d 2 -p cbr -e 1000 -c 0 -i 3 -a 0 -x {0.0 1.0 3 ----- null}
^t 1.00032 -s 0 -d 2 -p cbr -e 1000 -c 0 -i 4 -a 0 -x {0.0 1.0 4 ----- null}
h ^t 1.00032 -s 0 -d 2 -p cbr -e 1000 -c 0 -i 4 -a 0 -x {0.0 1.0 4 ----- null}
^t 1.00032 -s 0 -d 2 -p cbr -e 1000 -c 0 -i 4 -a 0 -x {0.0 1.0 4 ----- null}
h ^t 1.00032 -s 0 -d 2 -p cbr -e 1000 -c 0 -i 4 -a 0 -x {0.0 1.0 4 ----- null}
^t 1.0004 -s 0 -d 2 -p cbr -e 1000 -c 0 -i 5 -a 0 -x {0.0 1.0 5 ----- null}
h ^t 1.0004 -s 0 -d 2 -p cbr -e 1000 -c 0 -i 5 -a 0 -x {0.0 1.0 5 ----- null}
^t 1.0004 -s 0 -d 2 -p cbr -e 1000 -c 0 -i 5 -a 0 -x {0.0 1.0 5 ----- null}
h ^t 1.0004 -s 0 -d 2 -p cbr -e 1000 -c 0 -i 5 -a 0 -x {0.0 1.0 5 ----- null}
^t 1.00048 -s 0 -d 2 -p cbr -e 1000 -c 0 -i 6 -a 0 -x {0.0 1.0 6 ----- null}
h ^t 1.00048 -s 0 -d 2 -p cbr -e 1000 -c 0 -i 6 -a 0 -x {0.0 1.0 6 ----- null}
^t 1.00048 -s 0 -d 2 -p cbr -e 1000 -c 0 -i 6 -a 0 -x {0.0 1.0 6 ----- null}
h ^t 1.00048 -s 0 -d 2 -p cbr -e 1000 -c 0 -i 6 -a 0 -x {0.0 1.0 6 ----- null}
^t 1.00056 -s 0 -d 2 -p cbr -e 1000 -c 0 -i 7 -a 0 -x {0.0 1.0 7 ----- null}
h ^t 1.00056 -s 0 -d 2 -p cbr -e 1000 -c 0 -i 7 -a 0 -x {0.0 1.0 7 ----- null}
^t 1.00056 -s 0 -d 2 -p cbr -e 1000 -c 0 -i 7 -a 0 -x {0.0 1.0 7 ----- null}
h ^t 1.00056 -s 0 -d 2 -p cbr -e 1000 -c 0 -i 7 -a 0 -x {0.0 1.0 7 ----- null}
^t 1.00064 -s 0 -d 2 -p cbr -e 1000 -c 0 -i 8 -a 0 -x {0.0 1.0 8 ----- null}
h ^t 1.00064 -s 0 -d 2 -p cbr -e 1000 -c 0 -i 8 -a 0 -x {0.0 1.0 8 ----- null}
^t 1.00064 -s 0 -d 2 -p cbr -e 1000 -c 0 -i 8 -a 0 -x {0.0 1.0 8 ----- null}
h ^t 1.00064 -s 0 -d 2 -p cbr -e 1000 -c 0 -i 8 -a 0 -x {0.0 1.0 8 ----- null}
[ Read 899889 lines ]
^C Help      ^O Write Out  ^N Where Is   ^K Cut        ^T Execute    ^C Location   ^U Undo       ^M-A Set Mark  ^I To Bracket ^M-C Previous ^B Back
^X Exit      ^R Read File  ^X Replace    ^J Paste      ^_ Justify    ^G Go To Line ^E Redo       ^M-d Copy      ^G Where Mas  ^M-N Next     ^F Forward
```

NS2 Network Topology



Defining Network Links in NS2 TCL Script

```
#=====
#      Links Definition
#=====
#Createlinks between nodes
$ns duplex-link $n0 $n2 100.0Mb 10ms DropTail
$ns queue-limit $n0 $n2 50
$ns duplex-link $n2 $n1 99.9Mb 10ms DropTail
$ns queue-limit $n2 $n1 50

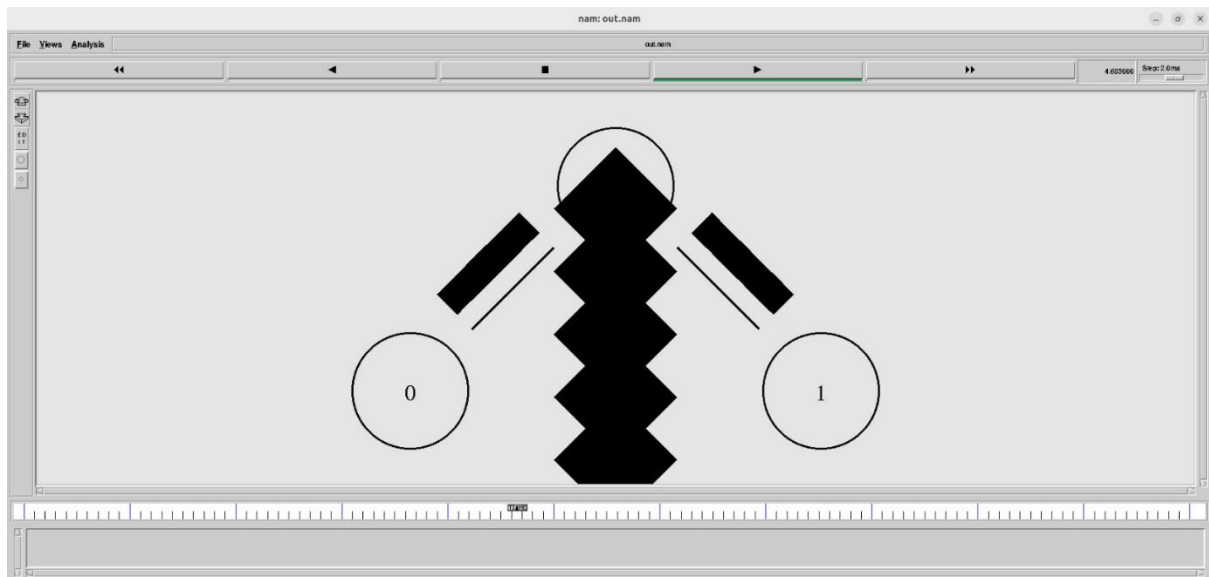
#Give node position (for NAM)
$ns duplex-link-op $n0 $n2 orient right-up
$ns duplex-link-op $n2 $n1 orient right-down
```

Packet Drop Count in NS2

```
sanchita@sanchita-VirtualBox:~/Downloads$ less out.tr | grep d | wc -l
64
sanchita@sanchita-VirtualBox:~/Downloads$
```

64 packets missed when i/p is 100.0mbps and o/p is 99.9mbps

NS2 Network Topology



Defining Network Links in NS2 TCL Script

```
#=====
#       Links Definition
#=====
#Createlinks between nodes
$ns duplex-link $n0 $n2 100.0Mb 10ms DropTail
$ns queue-limit $n0 $n2 50
$ns duplex-link $n2 $n1 99.0Mb 10ms DropTail
$ns queue-limit $n2 $n1 50

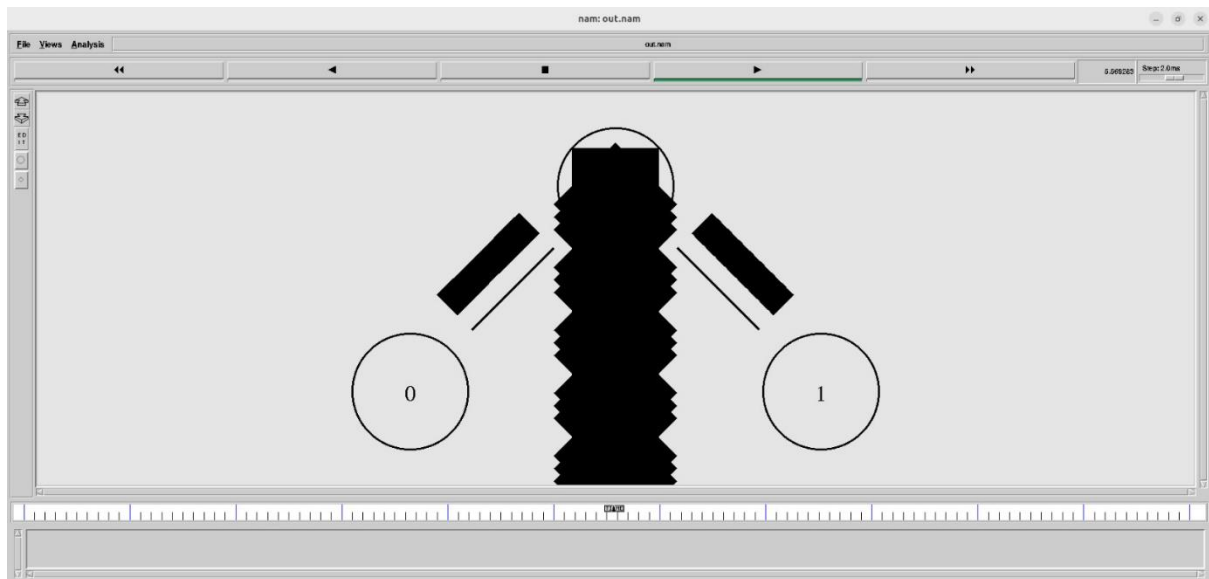
#Give node position (for NAM)
$ns duplex-link-op $n0 $n2 orient right-up
$ns duplex-link-op $n2 $n1 orient right-down
```

Packet Drop Count in NS2

```
sanchita@sanchita-VirtualBox:~/Downloads$ less out.tr | grep d | wc -l
1077
sanchita@sanchita-VirtualBox:~/Downloads$
```

1077 packets missed when i/p is 100.0mbps and o/p is 99.0mbps

NS2 Network Topology



Defining Network Links in NS2 TCL Script

```
#=====
#       Links Definition
#=====
#Createlinks between nodes
$ns duplex-link $n0 $n2 100.0Mb 10ms DropTail
$ns queue-limit $n0 $n2 50
$ns duplex-link $n2 $n1 95.0Mb 10ms DropTail
$ns queue-limit $n2 $n1 50

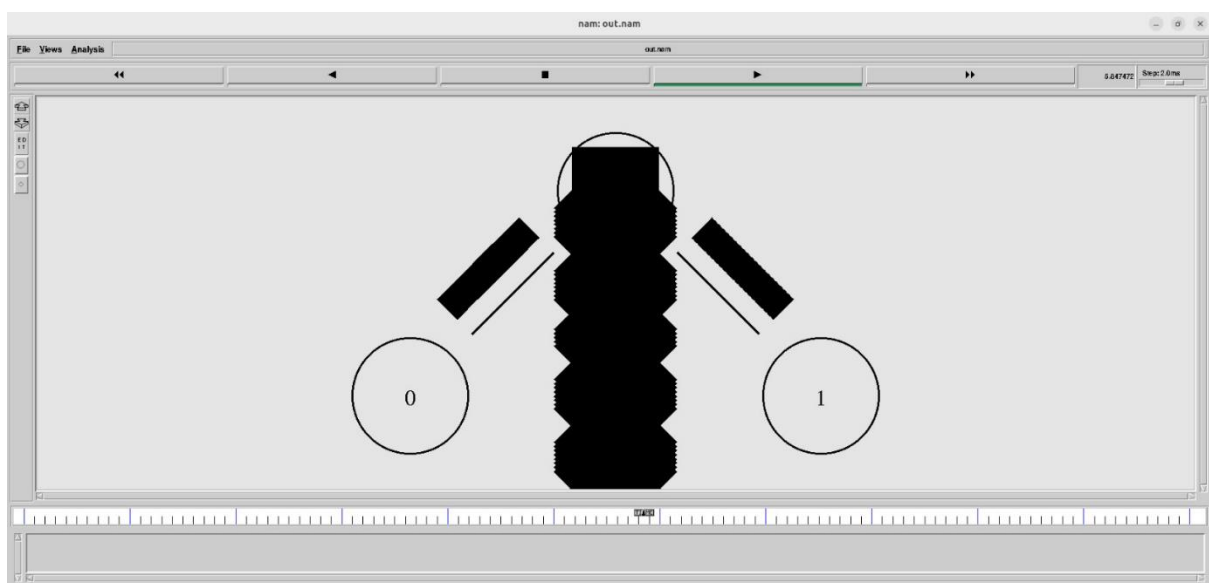
#Give node position (for NAM)
$ns duplex-link-op $n0 $n2 orient right-up
$ns duplex-link-op $n2 $n1 orient right-down
```

Packet Drop Count in NS2

```
sanchita@sanchita-VirtualBox:~/Downloads$ less out.tr | grep d | wc -l
5576
sanchita@sanchita-VirtualBox:~/Downloads$
```

5576 packets missed when i/p is 100.0mbps and o/p is 95.0mbps

NS2 Network Topology



Defining Network Links in NS2 TCL Script

```
#=====
#      Links Definition
#=====
#Createlinks between nodes
$ns duplex-link $n0 $n2 100.0Mb 10ms DropTail
$ns queue-limit $n0 $n2 50
$ns duplex-link $n2 $n1 85.0Mb 10ms DropTail
$ns queue-limit $n2 $n1 50

#Give node position (for NAM)
$ns duplex-link-op $n0 $n2 orient right-up
$ns duplex-link-op $n2 $n1 orient right-down
```

Packet Drop Count in NS2

```
sanchita@sanchita-VirtualBox:~/Downloads$ less out.tr | grep d | wc -l  
16827  
sanchita@sanchita-VirtualBox:~/Downloads$
```

16827 packets missed when i/p is 100.0mbps and o/p is 85.0mbps